Symmetry plays a central role in phase transitions. In almost all cases, if we compare a suitable measure of something's symmetry before and after it goes through a phase transition, we find a significant change. On a molecular scale, for instance, ice has a crystalline form with H_2O molecules arranged in an ordered, hexagonal lattice. Like the symmetries of the box in Figure 8.1, the overall pattern of the ice molecules is left unchanged only by certain special manipulations, such as rotations in units of 60 degrees about particular axes of the hexagonal arrangement. By contrast, when we heat ice, the crystalline arrangement melts into a jumbled, uniform clump of molecules—liquid water—that remains unchanged under rotations by any angle, about any axis. So, by heating ice and causing it to go through a solid-to-liquid phase transition, we have made it more symmetric. (Remember, although you might intuitively think that something more ordered, like ice, is more symmetric, quite the opposite is true; something is more symmetric if it can be subjected to more transformations, such as rotations, while its appearance remains unchanged.)

Similarly, if we heat liquid water and it turns into gaseous steam, the phase transition also results in an increase of symmetry. In a clump of water, the individual H_2O molecules are, on average, packed together with the hydrogen side of one molecule next to the oxygen side of its neighbor. If you were to rotate one or another molecule in a clump it would noticeably disrupt the molecular pattern. But when the water boils and turns into steam, the molecules flit here and there freely; there is no longer any pattern to the orientations of the H_2O molecules and hence, were you to rotate a molecule or group of molecules, the gas would look the same. Thus, just as the ice-to-water transition results in an increase in symmetry, the water-to-steam transition does so as well. Most (but not all[2]) substances behave in a similar way, experiencing an increase of symmetry when they undergo solid-to-liquid and liquid-to-gas phase transitions.

The story is much the same when you cool water or almost any other substance; it just takes place in reverse. For example, when you cool gaseous steam, at first not much happens, but as its temperature drops to 100 degrees Celsius, it suddenly starts to condense into liquid water; when you cool liquid water, not much happens until you reach 0 degrees